

# Exploratory Data Analysis on Airbnb Listings Dataset

## Introduction

Airbnb is one of the world's largest online accommodation marketplaces, connecting travellers with property owners across various locations. Understanding customer preferences, pricing behaviour, and listing patterns is important for hosts, travellers, and business analysts.

This project performs an in-depth Exploratory Data Analysis (EDA) on the New York City Airbnb Listings Dataset to uncover hidden trends, pricing patterns, neighbourhood insights, and customer behaviour using Python-based data analysis techniques.

**The analysis was carried out using:** Python, Pandas, NumPy, Matplotlib, Seaborn

## Project Objectives

**The primary objectives of this project were:**

To analyse Airbnb pricing trends across neighbourhoods

To identify the most preferred room types

To study how availability impacts listing prices

To detect outliers and anomalies in listing prices

To understand customer engagement through reviews

To generate meaningful business insights from the dataset

## Dataset Information

The dataset contains Airbnb listing information for New York City, including:

Listing price, Room type, Neighbourhood group, Availability, Number of reviews, Host information, Location details,

Dataset Size Rows: 48,895 ; Columns: 16

## Data Cleaning and Preprocessing

Before performing analysis, the dataset was cleaned and prepared.

## Steps Performed

- Checked for missing values
- Removed unnecessary columns:
  - last\_review
  - host\_name
  - name
- Verified duplicate records
- Handled missing values in reviews\_per\_month
- Checked incorrect data types
- Detected extreme price outliers

## Observations

The dataset contained missing values mainly in review-related columns.

No duplicate records were found.

Several listings contained unusually high prices, indicating possible luxury listings or data anomalies.

## Exploratory Data Analysis

### 1. Price Distribution Analysis

A histogram was used to analyse the distribution of Airbnb listing prices.

#### Key Observation

- The price distribution was highly right-skewed, showing that:
- Most Airbnb listings were priced below \$200
- A small number of luxury listings created extreme outliers
- This indicates that affordable and mid-range accommodations dominate the NYC Airbnb market.

### 2. Room Type Analysis

A count plot was used to analyse customer accommodation preferences.

#### Key Observation

- Entire homes/apartments were the most common room type, while shared rooms represented only a small percentage of total listings.
- This suggests that customers strongly prefer privacy and complete accommodation access.

### 3. Neighbourhood Analysis

Neighbourhood distribution was analysed using count plots and average pricing comparisons.

#### Key Observation

- Manhattan had the highest average listing prices
- Brooklyn contained a large number of listings at comparatively lower prices
- This shows that location significantly impacts Airbnb pricing behaviour.

### 4. Price vs Room Type

A boxplot was used to compare prices across different room types.

#### Key Observation

- Entire homes/apartments had significantly higher median prices
- Shared rooms had the lowest pricing distribution
- This confirms that room type is a major factor affecting Airbnb prices.

### 5. Availability vs Price

A scatter plot was created to study the relationship between listing availability and price.

#### Key Observation

No strong linear relationship was observed between availability and pricing. However, highly priced listings often showed lower booking frequency and reduced availability.

### 6. Correlation Analysis

A heatmap was generated to identify correlations between numerical features.

#### Key Observation

- Review count showed weak correlation with pricing
- Availability and reviews demonstrated moderate relationships
- Most variables showed low direct correlation, indicating that Airbnb pricing is influenced by multiple combined factors.

## Hypothesis Testing

**Null Hypothesis ( $H_0$ ) :** Room type does not significantly affect Airbnb prices.

**Alternative Hypothesis ( $H_1$ ) :** Room type significantly affects Airbnb prices.

A one-way **ANOVA** test was performed to determine whether room type significantly affects Airbnb prices.

**Result:** The p-value obtained from the statistical test was below the significance level of 0.05.

Therefore, the null hypothesis was rejected.

Room type has a statistically significant impact on Airbnb pricing.

## Key Business Insights

- Manhattan listings command premium prices compared to other boroughs.
- Entire homes/apartments dominate the Airbnb market.
- Most listings fall within affordable pricing ranges below \$200.
- Luxury listings create significant price outliers in the dataset.
- Shared rooms are the least preferred accommodation type.
- Some hosts manage multiple listings, indicating commercial hosting behaviour.
- Higher-priced listings generally receive fewer customer reviews.

## Conclusion

This project successfully uncovered major trends and pricing behaviours within the NYC Airbnb market using exploratory data analysis techniques.

### The analysis revealed that:

- Neighbourhood location strongly influences pricing
- Entire homes/apartments attract premium prices
- Most Airbnb listings remain budget-friendly
- Extreme price outliers significantly affect overall price distribution

### The project also demonstrated practical skills in:

Data cleaning, Data visualization, Statistical analysis, Business insight generation, Exploratory data analysis using Python

## Future Scope

### Future improvements for this project may include:

- Building machine learning models for price prediction
- Creating interactive dashboards using Power BI or Tableau
- Performing geospatial analysis using latitude and longitude data
- Conducting sentiment analysis on customer reviews
- Identifying seasonal pricing trends

### Tools & Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook